ON A GENUS 0 PROPER CURVE AN INVERTIBLE SHEAF OF POSITIVE DEGREE D HAS D+1 SECTIONS, VANISHING H^1, AND EMBEDS X INTO P^D

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the section sequence with its divisor, the trivialization argument on the Artinian zero scheme, the vanishing of H^1, the multiplication-map surjectivity with its two-part argument, and the Proj comparison closing the embedding are all retained verbatim, including the author's footnote and his remark that one verification is omitted.

Editorial note. The label lemma-genus-zero-positive-degree names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Effective Cartier divisors from sections. Divisors, Lemma divisors-lemma-characterize-OD and Remark divisors-remark-ses-regular-section. Supplies the short exact sequence 0 -> O_X -> L -> L|_D -> 0 attached to a nonzero section s with zero scheme D.
- Ampleness and Proj comparison on a curve. Varieties, Lemma varieties-lemma-ample-curve; Morphisms, Lemma morphisms-lemma-proper-ample-is-proj; Constructions, Lemma constructions-lemma-surjective-graded-rings-generated-degree-1-map-proj. Identifies X with Proj of the section ring and converts the surjectivity of the multiplication maps into the closed immersion into P^d_k.

Source reference. Original lemma 01NE: sources/stacks-chow/tags.txt, source lines 2155–2155. Prerequisite pointer invoked at proof line(s) 2103; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01N1: sources/stacks-chow/tags.txt, source lines 2142–2142. Prerequisite pointer invoked at proof line(s) 2100; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01X0: sources/stacks-chow/tags.txt, source lines 2501–2501. Prerequisite pointer invoked at proof line(s) 2042; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0BUT: sources/stacks-chow/tags.txt, source lines 14886–14886. Prerequisite pointer invoked at proof line(s) 2054; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0C6K: sources/stacks-chow/tags.txt, source lines 15300–15300. Prerequisite pointer invoked at proof line(s) 2043; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C6J: sources/stacks-chow/tags.txt, source lines 15299–15299. Prerequisite pointer invoked at proof line(s) 2086; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B5X: sources/stacks-chow/tags.txt, source lines 14050–14050. Prerequisite pointer invoked at proof line(s) 2084; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AYT: sources/stacks-chow/tags.txt, source lines 13801–13801. Prerequisite pointer invoked at proof line(s) 2048; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a proper curve over a field k with $H^0(X, \mathcal{O}_X) = k$. Assume X has genus 0. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module of degree $d > 0$. Then we have*

1. *$\dim_k H^0(X, \mathcal{L}) = d + 1$ and $\dim_k H^1(X, \mathcal{L}) = 0$,*
2. *$\mathcal{L}$ is very ample and defines a closed immersion into $\mathbf{P}^d_k$.*

Proof. By definition of degree and genus we have

$$\dim_k H^0(X, \mathcal{L}) - \dim_k H^1(X, \mathcal{L}) = d + 1$$

Let s be a nonzero section of $\mathcal{L}$. Then the zero scheme of s is an effective Cartier divisor $D \subset X$, we have $\mathcal{L} = \mathcal{O}_X(D)$ and we have a short exact sequence

$$0 \to \mathcal{O}_X \to \mathcal{L} \to \mathcal{L}|_D \to 0$$

see Divisors, Lemma 01X0 and Remark 0C6K. Since $H^1(X, \mathcal{O}_X) = 0$ by assumption, we see that $H^0(X, \mathcal{L}) \to H^0(X, \mathcal{L}|_D)$ is surjective. As $\mathcal{L}|_D$ is generated by global sections (because $\dim(D) = 0$, see Varieties, Lemma 0AYT) we conclude that the invertible module $\mathcal{L}$ is generated by global sections. In fact, since D is an Artinian scheme we have $\mathcal{L}|_D \cong \mathcal{O}_D$[1] and hence we can find a section t of $\mathcal{L}$ whose restriction of D generates $\mathcal{L}|_D$. The short exact sequence also shows that $H^1(X, \mathcal{L}) = 0$.

For $n \geq 1$ consider the multiplication map

$$\mu_n : H^0(X, \mathcal{L}) \otimes_k H^0(X, \mathcal{L}^{\otimes n}) \longrightarrow H^0(X, \mathcal{L}^{\otimes n+1})$$

We claim this is surjective. To see this we consider the short exact sequence

$$0 \to \mathcal{L}^{\otimes n} \xrightarrow{s} \mathcal{L}^{\otimes n+1} \to \mathcal{L}^{\otimes n+1}|_D \to 0$$

The sections of $\mathcal{L}^{\otimes n+1}$ coming from the left in this sequence are in the image of μ_n. On the other hand, since $H^0(\mathcal{L}) \to H^0(\mathcal{L}|_D)$ is surjective and since t^n maps to a trivialization of $\mathcal{L}^{\otimes n}|_D$ we see that $\mu_n(H^0(X, \mathcal{L}) \otimes t^n)$ gives a subspace of $H^0(X, \mathcal{L}^{\otimes n+1})$ surjecting onto the global sections of $\mathcal{L}^{\otimes n+1}|_D$. This proves the claim.

[1] In our case this follows from Divisors, Lemma 0BUT as $D \to \mathrm{Spec}(k)$ is finite.

Observe that $\mathcal{L}$ is ample by Varieties, Lemma 0B5X. Hence Morphisms, Lemma 0C6J gives an isomorphism

$$X \longrightarrow \mathrm{Proj}\left(\bigoplus\nolimits_{n\geq 0} H^0(X, \mathcal{L}^{\otimes n})\right)$$

Since the maps μ_n are surjective for all $n \geq 1$ we see that the graded algebra on the right hand side is a quotient of the symmetric algebra on $H^0(X, \mathcal{L})$. Choosing a k-basis $s_0, \ldots, s_d$ of $H^0(X, \mathcal{L})$ we see that it is a quotient of a polynomial algebra in $d+1$ variables. Since quotients of graded rings correspond to closed immersions of Proj (Constructions, Lemma 01N1) we find a closed immersion $X \to \mathbf{P}^d_k$. We omit the verification that this morphism is the morphism of Constructions, Lemma 01NE associated to the sections $s_0, \ldots, s_d$ of $\mathcal{L}$. $\square$